

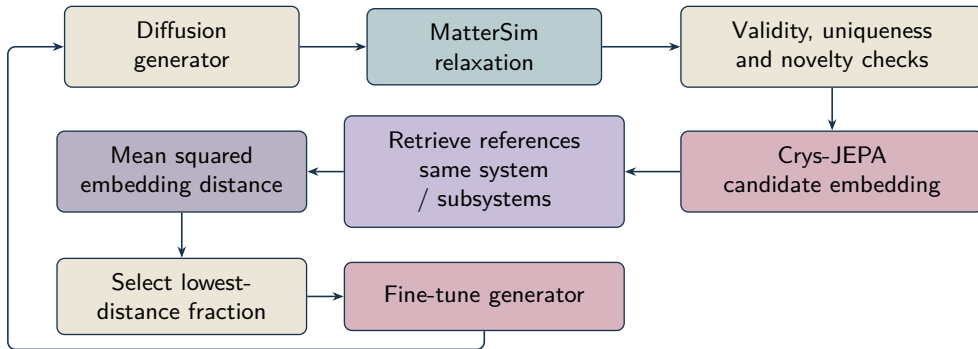
Reference-based screening and generation of crystals

Enriched reference pools and learned graph context

Screening generated candidates, followed by reference-conditioned generation.

The architectures and improvements discussed here are proposed,
not implemented or experimentally validated results.

Crys-JEPA: existing pipeline



- Crys-JEPA is the crystal encoder; the diffusion generator is a separate model.
- References are whole crystals. The refinement loop does not require new DFT calculations.

Existing method: Liu et al., Crys-JEPA [1]. Relaxation: MatterSim [4].

Crys-JEPA screening score

For a relaxed candidate c , retrieve training crystals whose elements lie within the candidate's chemical system:

$$R(c) = \{r \in \mathcal{D}_{\text{train}} : \text{elements}(r) \subseteq \text{elements}(c)\}.$$

The ranking score is the mean squared distance in Crys-JEPA embedding space:

$$s(c) = \frac{1}{|R(c)|} \sum_{r \in R(c)} \|h_c - h_r\|_2^2.$$

- For an AB_2 candidate, eligible references include A , B , and known A - B phases.
- Each retrieved reference contributes equally. Smaller scores rank first.
- The score measures proximity in an energy-aware representation; it is not an explicit energy-above-hull calculation (this is our understanding).

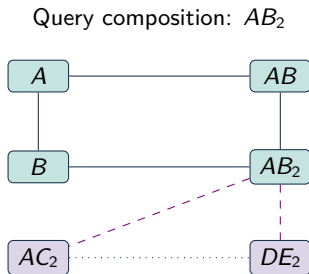
Existing scoring rule: Crys-JEPA [1].

Limitations of the reference pool

- **Chemical-system restriction.** The original pool excludes outside-system crystals that may contain transferable structural or chemical information.
- **Unequal information, equal weights.** References can differ in relevance and redundancy, but the score assigns equal weight to each.
- **No explicit joint competition.** Pairwise distances do not directly represent composition-balanced alternatives such as $AB + B$ for AB_2 .
- **Reference-set dependence.** Adding references changes both the centroid and spread of their embeddings, and therefore the average-distance score.

These observations motivate experiments. We are claiming theoretically that this can be improved.

Proposed reference graph

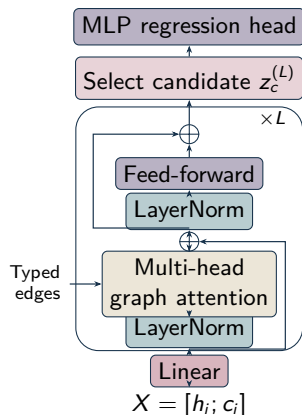


Each node is a whole crystal

- **Competition references:** crystals using the candidate's elements.
- **Cross-chemistry analogues:** crystals with relevant structural or chemical similarities.
- Retrieve a sparse neighbourhood from a shared graph, then attach the generated candidate as a query node.
- Define retrieval and edge rules explicitly before learning graph aggregation.

Analogues supply predictive context. They cannot be decomposition products when they introduce elements absent from the candidate.

Candidate screening with a graph transformer



Relation-aware graph transformer

- Inputs: frozen Crys-JEPA embeddings, composition features and typed graph edges.
- Multi-head attention exchanges information between connected crystal nodes.
- LayerNorm, feed-forward layers and residual connections update the node representations.

Separate MLP regression head

- Select the final candidate vector $z_c^{(L)}$.
- A small feed-forward network maps it to one predicted energy-above-hull value.

$$\widehat{E}_{\text{hull}}(c) = \text{MLP}_{\text{head}}(z_c^{(L)}).$$

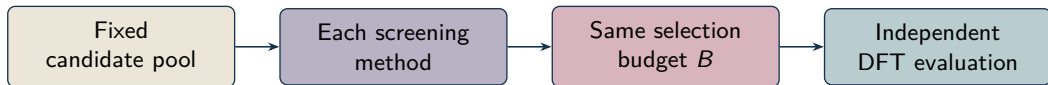
Known energy-above-hull values supply training labels; they are not candidate input features.

Screening evaluation

Comparison	Question
Original score; candidate-only predictor	Do references improve ranking over the existing score and candidate features alone?
Nearest-neighbour / simple aggregation; matched set-attention model	Is learned aggregation sufficient without graph structure?
Restricted-reference vs. enriched-reference GNN	Do cross-chemistry analogues add useful information?
Remove reference-reference edges; perturb edge types	Do the relationships themselves contribute?
Energy predictor + conventional hull evaluation	Does the graph offer value over a direct energy-based approach?

- Fix the generated candidate pool, selection budget and available supervision.
- Match reference features and counts for graph/set comparisons. Report coverage and runtime.
- Hold out related structures together; test transfer to unseen chemical systems.

Screening validation



- Primary outcome: number and fraction of selected candidates below a predefined E_{hull} threshold.
- Use compatible energy methods and corrections, and sufficient competition coverage for hull evaluation.
- If only a subset receives DFT, use the same prespecified sampling procedure for each method.
- Report failed relaxations and calculations. Keep learned selection scores separate from final validation labels.

A near-hull threshold indicates thermodynamic competitiveness under the evaluation protocol, not demonstrated synthesizability.

Using reference context during generation

The same reference database graph may also inform how a candidate is constructed.

Selection-based refinement

- Score completed candidates.
- Select examples for generator fine-tuning.
- References affect the generator indirectly through training data.

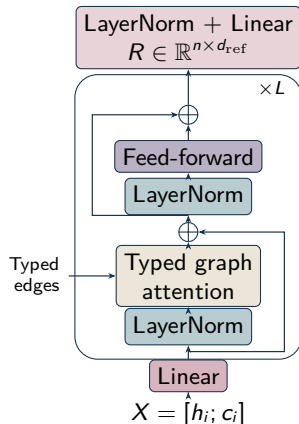
Proposed direct conditioning

- Retrieve references before generation.
- Encode them as contextual tokens.
- Let the denoiser consult these tokens through cross-attention.

The initial scope chooses a target composition before generation. Any improvement over composition-only conditioning must be established experimentally.

Composable Crystals motivates reusable structural conditioning [2]; MatterGen provides diffusion-generation context [3].

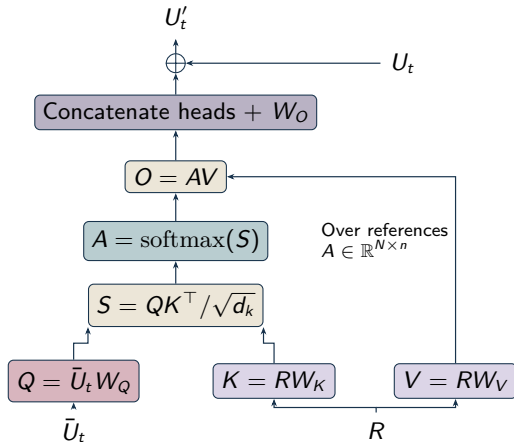
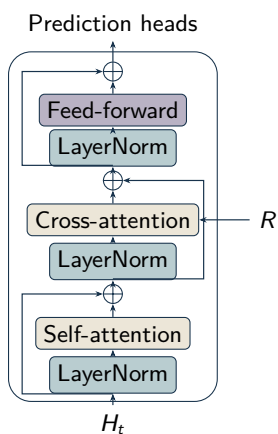
Reference graph encoder



- Input: frozen Crys-JEPA embeddings and composition features for retrieved references.
- Typed graph attention contextualizes each crystal using its neighbours.
- Output: one vector per reference, $R \in \mathbb{R}^{n \times d_{\text{ref}}}$; no global pooling.
- The generation GNN learns through the denoising loss. Its weights initially differ from the screening GNN's.

Whole-crystal tokens provide compressed context. They do not directly encode transferable local atomic geometries.

Reference cross-attention



Candidate states supply queries. Reference tokens supply keys and values. Attention selects reference information and adds it to the candidate state.

Training and inference

Training

1. Choose a labelled training crystal and its composition.
2. Retrieve references using composition; exclude the target and equivalent duplicates.
3. Encode the reference subgraph.
4. Corrupt the training crystal at a sampled diffusion timestep.
5. Optimize the standard denoising loss through the graph encoder, cross-attention and selected denoiser weights.

Retrieval during training must use only information available at inference. Inference does not update weights; selected outputs may support a later refinement round.

Inference

1. Specify the requested composition.
2. Retrieve and encode references once.
3. Sample from the diffusion prior and denoise with reference conditioning.
4. Relax the result and apply validity, uniqueness and novelty checks.
5. Retrieve using the completed structure, screen, and select candidates.

Generation evaluation and design choices

Factorial comparison

Generator	Screening
Baseline	Baseline
Baseline	Proposed
Reference-conditioned	Baseline
Reference-conditioned	Proposed

- Include composition-only conditioning and unordered-reference controls.
- Compare relevant and shuffled references; vary reference count.
- Match training data, target compositions, sample counts and refinement budgets.

Open design choices

- Fix atom identities/counts, or condition atom-type diffusion on composition.
- Select graph edges, retrieval budget and graph-encoder depth.
- Choose which denoiser weights to fine-tune.
- Consider balanced mixture nodes only if initial screening results motivate them.

Validation of generated crystals

- **Before selection:** measure relaxation success, validity, uniqueness, novelty, diversity and chemical coverage over the full sampled pool.
- **After selection:** independently evaluate the same budget from each generator/screener combination using compatible DFT and hull protocols.
- **Discovery yield:** report independently validated stable or near-hull, unique, novel candidates per generated sample, alongside yield per validated candidate.
- **Reference dependence:** measure similarity to retrieved crystals and control near-duplicates across data splits.
- **Cost and failures:** include generation, retrieval, screening and validation overhead; report unsuccessful calculations.

Improvement requires higher independently validated yield, not only greater agreement with the learned screening model.

References

- [1] Nian Liu et al.
Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement.
2026. [arXiv:2605.14759](#).
- [2] Nian Liu et al.
Composable Crystals: Controllable Materials Discovery via Concept Learning.
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- [3] Claudio Zeni et al.
A generative model for inorganic materials design.
Nature, 2025. doi:10.1038/s41586-025-08628-5.
- [4] Han Yang et al.
MatterSim: A Deep Learning Atomistic Model Across Elements, Temperatures and Pressures.
2024. [arXiv:2405.04967](#).

Questions?